

Congenital rubella deafness: a preventable disease.

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Over a 4-year period (1972-75) an unselected sample of 568 children aged under 4 years attending the Nuffield Hearing and Speech Centre were tested for rubella antibody. Sensorineural deafness was subsequently diagnosed in 349 of these children, and 83 (24%) of this group had rubella antibody. In contrast, only 19 (9%) of the remaining 219 children in whom sensorineural deafness was excluded had rubella antibody. The seropositive and seronegative children with sensorineural deafness showed striking differences in family history, history of maternal rubella, adverse perinatal events, and presence of other defects. Congenital rubella is an important cause of deafness, and the rubella vaccination programme must be pursued more vigorously if this serious defect is to be prevented in the future.

Congenital Cytomegalovirus and Deafness.

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Although the incidence of medical and neurological problems resulting from congenital cytomegalovirus (CMV) infection is relatively low, the widespread and indiscriminate nature of this infection and the severity of these conditions when they do occur are such that it warrants the close attention of medical specialists, audiologists, and educators. The identification of congenital CMV is especially difficult because of its largely nonsymptomatic character, and because conditions associated with it, including hearing impairment, can be either progressive in nature or occur only later in life. Data reviewed in this study resemble those reported for children with impaired hearing from the 1964-65 maternal rubella epidemic: hearing loss in the severe to profound range, often accompanied by serious additional disabilities, especially mental retardation and cerebral palsy. Depressed achievement test results of children with CMV-induced hearing loss are further indications of the serious nature of this disease. The presence of any symptoms of CMV infection in infants or of risk factors associated with it-e.g., purplish skin rash, severe asphyxia, jaundice, low birth weight, swollen lymph glands, and other mononucleosis-like symptoms-signals the need for immediate testing, including audiological evaluation, and, if results are positive, the initiation of early medical and educational intervention.

Clinical profile of hearing loss in children with congenital cytomegalovirus (CMV) infection: CMV DNA diagnosis using preserved umbilical cord.

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Congenital cytomegalovirus (CMV) infection is a major cause of bilateral and unilateral sensorineural hearing loss (SNHL) in children, accounting for 9.0% of SNHL cases. The diagnostic rate using combined genetic deafness test and CMV DNA detection test was determined to be 46.4% in bilateral profound SNHL. The present study investigated the prevalence of congenital CMV infection diagnosed retrospectively by detection of CMV DNA in dried umbilical cord specimens from children with unilateral or bilateral SNHL up to the age of 12 years. Preserved dried umbilical cords were collected from 134 children with bilateral (46 children) or unilateral (88 children) SNHL. DNA was extracted from the dried umbilical cords and CMV DNA was detected by quantitative PCR. Genetic deafness tests based on the invader assay were performed in children with bilateral SNHL. CMV DNA from the dried umbilical cords

was detected in 8.7% of the bilateral SNHL and 9.1% of unilateral SNHL. Deafness gene mutations were identified in 21.7% (10/46) of children with bilateral SNHL.

[The very severe sensorineural deafness patients caused by rubella virus infection: two cases report].

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To explore the audiological features in children who were severe sensorineural hearing loss infected with rubella virus. There were two cases of rubella virus infection in children who were deaf, they conducted the distortion product otoacoustic emission, ABR and auditory steady-state evoked response (ASSR) examination, then analyzed the results comprehensively. Two patients' mothers were prompted to have infected rubella virus during the early three months pregnant period by history and laboratory tests. The two patients were not detected deafness gene mutation. Audiology results implied the two patients were very severe bilateral sensorineural deafness, so they were recommended to be equipped with hearing aids and cochlear implant surgery. Early pregnancy women infected with rubella virus can cause very severe offspring sensorineural deafness. The crowd whose mother were suspected to infect with rubella virus in early pregnancy, that should be tracked and detected hearing in order to achieve early detection, early intervention and early treatment.

Analysis of selective screening for congenital cytomegalovirus in a secondary hospital: Problems and solutions.

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Universal hearing screening offers unique possibilities for detection of congenital deafness as a consequence of congenital cytomegalovirus (CMVc) infection, so its selective study in the case of a failed test could be a non-negligible screening opportunity while other guidelines covering the possibility of universal screening are adopted. The aim of this study is to analyse the possibility of selective screening for CMVc after an altered hearing test in a regional hospital. During the period studied, the results obtained were unsatisfactory, especially in children born outside the hospital of residence, showing an excessive delay in hearing screening in many cases and in the few cases where CMVc screening could be performed, only 30% had the test ordered in a timely manner. The reasons for this are varied and the solution is to include selective screening for CMVc in the hearing screening programme. This implies shortening the timing of the hearing screening protocol to allow CMVc testing in saliva or urine (preferably) before 21 days of age and providing screening programmes with the necessary staff and time to perform it properly.

Analysis of the Results of Cytomegalovirus Testing Combined with Genetic Testing in Children with Congenital Hearing Loss.

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To improve the etiological diagnosis of congenital hearing loss by combining whole-exome sequencing (WES) with cytomegalovirus (CMV) testing and to explore the potential benefits of adding CMV screening to newborn hearing screening, 80 children under 2 years of age with bilateral sensorineural hearing loss were recruited. Peripheral venous blood was extracted from the children for WES analysis. Saliva after mouthwash and the first urine in the morning were collected and used as samples to quantify CMV DNA copy number in urine and saliva by qPCR; among the 80 children with congenital deafness, 59 (74%) were found to have genetic variants that may cause congenital deafness, including 44 with GJB2 or SLC26A4 gene variant, 1 with STRC gene variant, and 14 with other genetic variants. A total of 12 children carried deafness gene variants associated with a syndrome; CMV test results showed that in two children, the CMV DNA copy number in saliva was $>1000/\text{mL}$, which indicates that they were CMV-positive, and their genetic test results were negative. A neonatal CMV test combined with genetic screening can improve the etiological diagnosis rate of congenital deafness, and the direct evidence of neonatal CMV infection deserves further verification.

Notes from the Field: Zika Virus-Associated Neonatal Birth Defects Surveillance - Texas, January 2016-July 2017.

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On November 28, 2016, the Texas Department of State Health Services (Texas DSHS) reported its first confirmed case of local mosquitoborne Zika virus transmission in the city of Brownsville, located in south Texas along the U.S.-Mexico border. Zika virus infection during pregnancy has been linked to adverse congenital outcomes including microcephaly, neural tube defects, early brain malformations, structural eye abnormalities, congenital deafness, and limb contractures (1). On January 1, 2016, Texas DSHS established enhanced surveillance to identify women with laboratory evidence of possible Zika virus infection during pregnancy and suspected cases of Zika virus-associated birth defects among completed pregnancies.

Cochlear implantation in children with congenital cytomegalovirus infection.

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To assess the impact of cochlear implantation (CI) on children with cytomegalovirus (CMV)-induced deafness. Retrospective chart review. Four children with congenital CMV-related deafness (CMV group) and 17 children with congenital deafness without CMV infection as the cause of deafness (non-CMV group). The age at CI ranged from 2.0 to 3.3 years (mean, 2.6 yr) in the CMV group and from 1.8 to 3.6 years (mean, 2.6 yr) in the non-CMV group. Their follow-up period ranged from 3.0 to 4.3 years (mean, 3.3 yr) in the CMV group and from 1.6 to 4.3 years (mean, 3.3 yr) in the non-CMV group. Evaluation and comparison of preoperative and postoperative hearing levels, motor, social, and language development, Infant-Toddler Meaningful Auditory Integration Scale, and Enjoji Scale of Infant Analytical Development between the 2 groups. Within 12 months after CI, the mean score of both language perception and production was poorer in the CMV group than in the non-CMV group, and the difference in the language production was statistically significant. However, 12 months after CI, the language perception and production showed good progress at levels similar to that of the non-CMV group. Long-term results of the language perception and production after CI was satisfactory in Japanese children with congenital

CMV-related deafness compared with in subjects deafened by other causes. CMV alone, as a cause of deafness, is not likely a contraindication for CI.

Diagnosis of human congenital cytomegalovirus infection by amplification of viral DNA from dried blood spots on perinatal cards.

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Congenital human cytomegalovirus (HCMV) infection affects 1% of children and is the most common infectious cause of sensorineural hearing loss. Due to the difficulty of diagnosing deafness and other neurological disorders in infants, affected individuals may not be recognized until much later when active infection has resolved and culture is no longer informative. To overcome this problem, congenital HCMV infection was diagnosed retrospectively by testing residual blood samples collected from newborns and dried on perinatal cards as part of the North Carolina Newborn Screening Program. We modified the Qiagen method for purifying DNA from dried blood spots to increase the sample size and recovery of the lysate. A multiplex, real-time TaqMan polymerase chain reaction assay on an ABI 7900 instrument measured a highly conserved segment of the HCMV polymerase gene and the APOB human control gene. HCMV DNA was detected in blood dried on perinatal cards from all seven infants with culture-proven congenital infection, and all 24 negative control cases lacked detectable HCMV DNA. Our findings suggest that it is possible to diagnose congenital HCMV infection using dried blood collected up to 20 months earlier. Further studies are warranted on patients with hearing loss or other neurological deficits to determine the percentage that is attributable to congenital HCMV infection.

Cytomegalovirus infection: perinatal implications.

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Cytomegalovirus (CMV), a member of the herpes virus family, is the most common cause of congenital infection in humans, affecting 0.5-3% of all newborns worldwide. Congenital cytomegalovirus infection is the leading infectious cause of deafness, learning disabilities, and mental retardation in children. The high prevalence of cytomegalovirus in the general population, unpredictability of transmission, and asymptomatic nature of the disease in otherwise healthy women challenge prevention and treatment efforts.
